

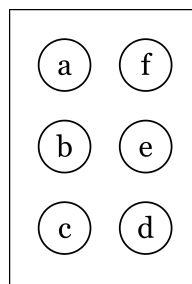
Complex Numbers 2, group *Alpha* α -i

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|8 - 4i|$??? $\sqrt{80}$
 (b) $1 - i$??? $\frac{7\pi}{4}$
 (c) $2\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? no
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $6e^{i\frac{11\pi}{6}}$??? $-3\sqrt{3} - 3i$
 (f) $x^4 = -2$??? $2i$

1.



Braille letter

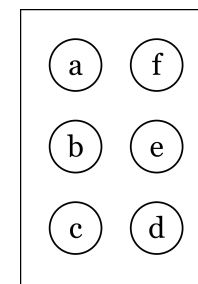
Complex Numbers 2, group *Alpha* α -ii

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 5i|$??? $\sqrt{26}$
 (b) $-3 - 3i$??? $\frac{7\pi}{4}$
 (c) $2(\cos 2\pi + i \sin 2\pi)$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $3e^{i\frac{\pi}{6}}$??? $\frac{3\sqrt{3}}{2} - \frac{3}{2}i$
 (f) $x^6 = -2$??? -2

2.



Braille letter

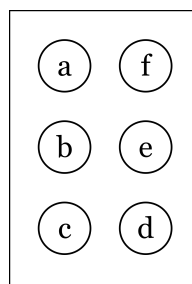
Complex Numbers 2, group *Alpha* α -iii

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|6 - i|$??? $\sqrt{37}$
 (b) $1 + i$??? $\frac{5\pi}{4}$
 (c) $8\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $e^{i\frac{5\pi}{6}}$??? $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
 (f) $x^5 = 4$??? 4

3.



Braille letter

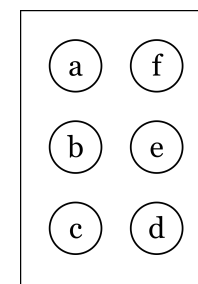
Complex Numbers 2, group *Alpha* α -iv

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|6 - 4i|$??? $\sqrt{52}$
 (b) $-4 + 4i$??? $\frac{5\pi}{4}$
 (c) $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $9e^{i\frac{2\pi}{3}}$??? $-9 + 9\sqrt{3}i$
 (f) $x^3 = 3$??? -3

4.



Braille letter

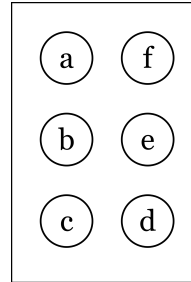
Complex Numbers 2, group *Beta* β -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-5 + 4i|$??? $\sqrt{42}$
 (b) $-2 - 2i$??? $\frac{5\pi}{4}$
 (c) $5\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $3e^{i\frac{\pi}{6}}$??? $-\frac{3\sqrt{3}}{2} + \frac{3}{2}i$
 (f) $x^5 = -2i$??? $-2i$

1.



Braille letter

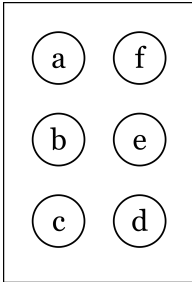
Complex Numbers 2, group *Beta* β -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-8 + 3i|$??? $\sqrt{73}$
 (b) $-1 + i$??? $\frac{3\pi}{4}$
 (c) $8\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$??? no
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $5e^{i\frac{4\pi}{3}}$??? $-\frac{5}{2} - \frac{5\sqrt{3}}{2}i$
 (f) $x^3 = 5i$??? $-5i$

2.



Braille letter

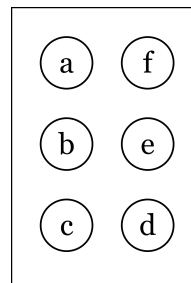
Complex Numbers 2, group *Beta* β -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|6 - 3i|$??? $\sqrt{45}$
 (b) $2 - 2i$??? $\frac{\pi}{4}$
 (c) $2\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $2e^{i\frac{7\pi}{6}}$??? $-\sqrt{3} - i$
 (f) $x^3 = -5i$??? -5

3.



Braille letter

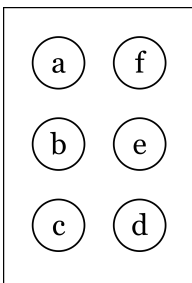
Complex Numbers 2, group *Beta* β -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 3i|$??? 5
 (b) $-5 - 5i$??? $\frac{5\pi}{4}$
 (c) $6\left(\cos \frac{7\pi}{2} + i \sin \frac{7\pi}{2}\right)$... ??? ... yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $e^{i\frac{5\pi}{3}}$??? $1 - \sqrt{3}i$
 (f) $x^3 = 4i$??? $4i$

4.



Braille letter

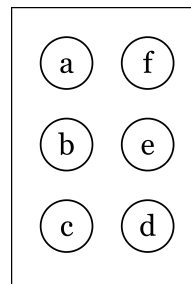
Complex Numbers 2, group *Gamma* γ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 2i|$??? 2
 (b) $-3 - 3i$??? $\frac{5\pi}{4}$
 (c) $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $5e^{i\frac{11\pi}{6}}$??? $\frac{5\sqrt{3}}{2} + \frac{5}{2}i$
 (f) $x^5 = 3$??? 3

1.



Braille letter

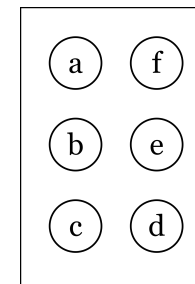
Complex Numbers 2, group *Gamma* γ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-5 + i|$??? $\sqrt{26}$
 (b) $-2 + 2i$??? $\frac{5\pi}{4}$
 (c) $7\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $e^{i\frac{\pi}{3}}$??? $1 + \sqrt{3}i$
 (f) $x^4 = -3$??? $3i$

2.



Braille letter

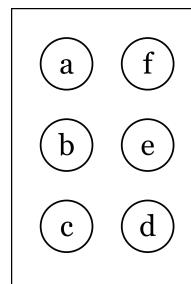
Complex Numbers 2, group *Gamma* γ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 2i|$??? $\sqrt{21}$
 (b) $-4 - 4i$??? $\frac{5\pi}{4}$
 (c) $7\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $2e^{i\frac{7\pi}{6}}$??? $-\sqrt{3} + i$
 (f) $x^4 = 9$??? -9

3.



Braille letter

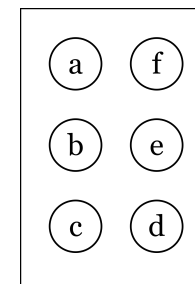
Complex Numbers 2, group *Gamma* γ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|7 - 5i|$??? 2
 (b) $-3 + 3i$??? $\frac{3\pi}{4}$
 (c) $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $2e^{i\frac{\pi}{6}}$??? $\sqrt{3} + i$
 (f) $x^6 = -2i$??? $2i$

4.



Braille letter

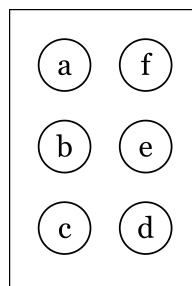
Complex Numbers 2, group *Delta* δ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|8 - 2i|$??? $\sqrt{69}$
 (b) $-3 + 3i$??? $\frac{3\pi}{4}$
 (c) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $3e^{i\frac{4\pi}{3}}$??? $-3 - 3\sqrt{3}i$
 (f) $x^4 = -4i$??? $4i$

1.



Braille letter

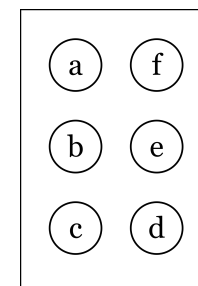
Complex Numbers 2, group *Delta* δ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 6i|$??? 5
 (b) $-1 + i$??? $\frac{3\pi}{4}$
 (c) $4(\cos 2\pi + i \sin 2\pi)$. ??? . yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $e^{i\frac{4\pi}{3}}$??? $-1 - \sqrt{3}i$
 (f) $x^5 = 8i$??? $8i$

2.



Braille letter

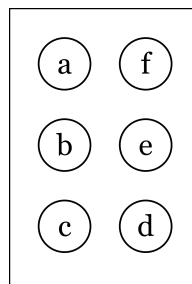
Complex Numbers 2, group *Delta* δ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 3i|$??? 5
 (b) $-2 + 2i$??? $\frac{5\pi}{4}$
 (c) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $e^{i\frac{5\pi}{4}}$??? $-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$
 (f) $x^4 = 8$??? 8

3.



Braille letter

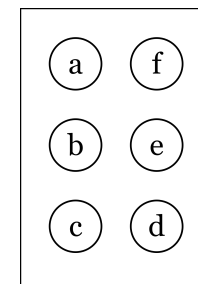
Complex Numbers 2, group *Delta* δ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|7 - 5i|$??? $\sqrt{74}$
 (b) $3 + 3i$??? $\frac{5\pi}{4}$
 (c) $4(\cos 3\pi + i \sin 3\pi)$... ??? ... yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $5e^{i\frac{4\pi}{3}}$??? $-\frac{5}{2} - \frac{5\sqrt{3}}{2}i$
 (f) $x^3 = -9i$??? -9

4.



Braille letter

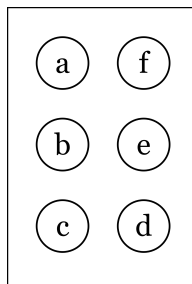
Complex Numbers 2, group *Epsilon* € -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + i|$??? $\sqrt{5}$
 (b) $4 - 4i$??? $\frac{\pi}{4}$
 (c) $7(\cos \pi + i \sin \pi)$??? no
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
 (e) $3e^{i\frac{\pi}{4}}$??? $\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$
 (f) $x^6 = 6$??? -6

1.



Braille letter

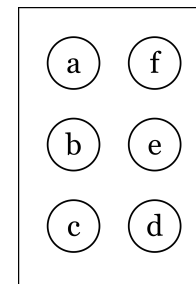
Complex Numbers 2, group *Epsilon* € -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + i|$??? $\sqrt{37}$
 (b) $-2 - 2i$??? $\frac{7\pi}{4}$
 (c) $2(\cos \pi + i \sin \pi)$??? no
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3})$
 (e) $2e^{i\frac{5\pi}{4}}$??? $-\sqrt{2} - \sqrt{2}i$
 (f) $x^4 = -2$??? $2i$

2.



Braille letter

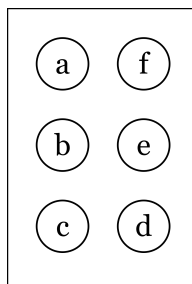
Complex Numbers 2, group *Epsilon* € -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 3i|$??? $\sqrt{11}$
 (b) $-3 + 3i$??? $\frac{3\pi}{4}$
 (c) $7(\cos \frac{9\pi}{2} + i \sin \frac{9\pi}{2})$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
 (e) $6e^{i\frac{5\pi}{6}}$??? $-3\sqrt{3} + 3i$
 (f) $x^4 = 4i$??? $-4i$

3.



Braille letter

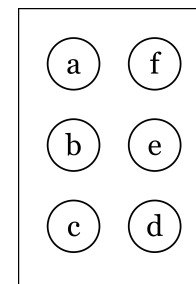
Complex Numbers 2, group *Epsilon* € -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-9 + 3i|$??? $\sqrt{90}$
 (b) $3 - 3i$??? $\frac{\pi}{4}$
 (c) $2(\cos 2\pi + i \sin 2\pi)$... ??? ... yes
 (d) $-1 + \sqrt{3}i$??? $2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$
 (e) $e^{i\frac{5\pi}{6}}$??? $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
 (f) $x^6 = 5$??? 5

4.



Braille letter

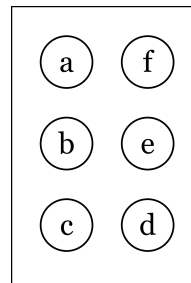
Complex Numbers 2, group *Zeta* ζ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 4i|$??? $\sqrt{41}$
 (b) $-5 + 5i$??? $\frac{5\pi}{4}$
 (c) $3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $5e^{i\frac{5\pi}{6}}$??? $\frac{5\sqrt{3}}{2} + \frac{5}{2}i$
 (f) $x^5 = 2$??? 2

1.



Braille letter

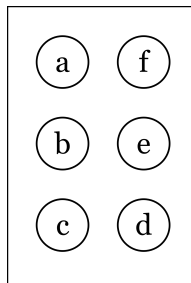
Complex Numbers 2, group *Zeta* ζ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 2i|$??? $\sqrt{21}$
 (b) $7 - 7i$??? $\frac{7\pi}{4}$
 (c) $4\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$??? no
 (d) $-3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
 (e) $8e^{i\frac{4\pi}{3}}$??? $-8 - 8\sqrt{3}i$
 (f) $x^4 = 2$??? -2

2.



Braille letter

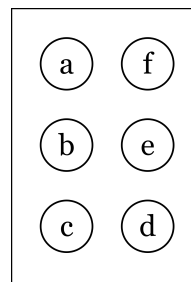
Complex Numbers 2, group *Zeta* ζ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|2 - 4i|$??? $\sqrt{20}$
 (b) $5 - 5i$??? $\frac{\pi}{4}$
 (c) $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $e^{i\frac{5\pi}{6}}$??? $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
 (f) $x^4 = -4$??? 4

3.



Braille letter

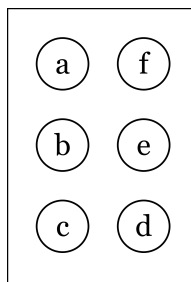
Complex Numbers 2, group *Zeta* ζ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 7i|$??? $\sqrt{65}$
 (b) $-7 + 7i$??? $\frac{5\pi}{4}$
 (c) $3\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... no
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $8e^{i\frac{2\pi}{3}}$??? $-4 + 4\sqrt{3}i$
 (f) $x^3 = -3$??? -3

4.



Braille letter

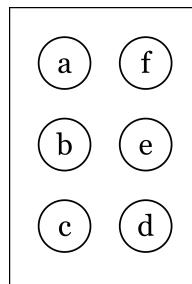
Complex Numbers 2, group *Eta* η -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 3i|$??? 1
 (b) $-4 - 4i$??? $\frac{5\pi}{4}$
 (c) $4(\cos \pi + i \sin \pi)$... ??? ... no
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $2e^{i\frac{7\pi}{4}}$??? $-\sqrt{2} - \sqrt{2}i$
 (f) $x^3 = 6i$??? $6i$

1.



Braille letter

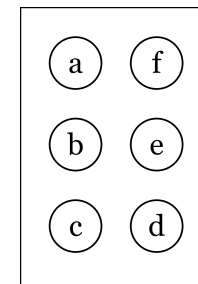
Complex Numbers 2, group *Eta* η -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-7 + 9i|$??? $\sqrt{131}$
 (b) $-2 + 2i$??? $\frac{3\pi}{4}$
 (c) $2(\cos \pi + i \sin \pi)$??? no
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $2e^{i\frac{\pi}{3}}$??? $1 + \sqrt{3}i$
 (f) $x^5 = 5i$??? $5i$

2.



Braille letter

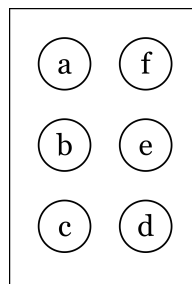
Complex Numbers 2, group *Eta* η -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|3 - i|$??? $\sqrt{10}$
 (b) $-5 + 5i$??? $\frac{5\pi}{4}$
 (c) $7(\cos 4\pi + i \sin 4\pi)$... ??? ... yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $2e^{i\frac{5\pi}{3}}$??? $-1 - \sqrt{3}i$
 (f) $x^6 = 7i$??? -7

3.



Braille letter

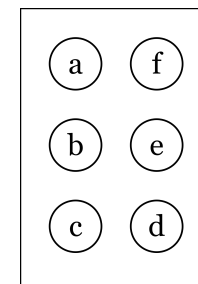
Complex Numbers 2, group *Eta* η -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 7i|$??? $\sqrt{65}$
 (b) $-8 + 8i$??? $\frac{3\pi}{4}$
 (c) $7(\cos \pi + i \sin \pi)$??? no
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $3e^{i\frac{4\pi}{3}}$??? $-\frac{3}{2} - \frac{3\sqrt{3}}{2}i$
 (f) $x^4 = 6$??? 6

4.



Braille letter

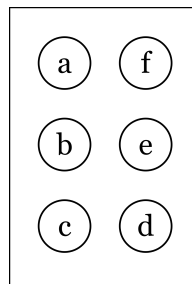
Complex Numbers 2, group *Theta* θ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-3 + 8i|$??? $\sqrt{73}$
 (b) $-3 - 3i$??? $\frac{5\pi}{4}$
 (c) $8(\cos 4\pi + i \sin 4\pi)$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
 (e) $3e^{i\frac{\pi}{4}}$??? $\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$
 (f) $x^5 = -7$??? -7

1.



Braille letter

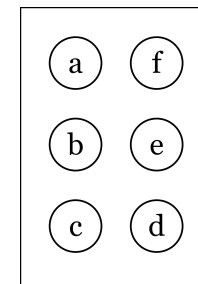
Complex Numbers 2, group *Theta* θ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|7 - 3i|$??? $\sqrt{59}$
 (b) $-4 + 4i$??? $\frac{3\pi}{4}$
 (c) $5(\cos \pi + i \sin \pi)$??? yes
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3})$
 (e) $3e^{i\frac{\pi}{6}}$??? $\frac{3\sqrt{3}}{2} - \frac{3}{2}i$
 (f) $x^3 = -5$??? -5

2.



Braille letter

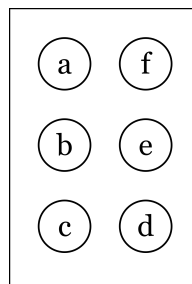
Complex Numbers 2, group *Theta* θ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 4i|$??? $\sqrt{17}$
 (b) $-5 + 5i$??? $\frac{3\pi}{4}$
 (c) $6(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2})$??? ... no
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3})$
 (e) $7e^{i\frac{\pi}{4}}$??? $7\sqrt{2} + 7\sqrt{2}i$
 (f) $x^3 = -5$??? -5

3.



Braille letter

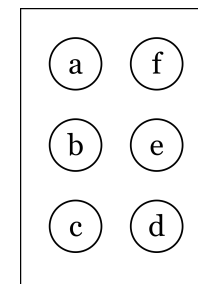
Complex Numbers 2, group *Theta* θ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 4i|$??? 3
 (b) $2 - 2i$??? $\frac{7\pi}{4}$
 (c) $5(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})$... ??? ... yes
 (d) $1 - \sqrt{3}i$??? $2(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6})$
 (e) $2e^{i\frac{\pi}{4}}$??? $\sqrt{2} + \sqrt{2}i$
 (f) $x^4 = -4i$??? $4i$

4.



Braille letter

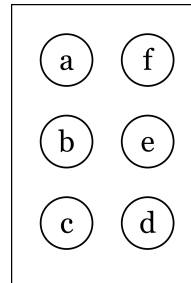
Complex Numbers 2, group *Iota* ι -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - i|$??? $\sqrt{26}$
 (b) $-5 + 5i$??? $\frac{5\pi}{4}$
 (c) $3(\cos 4\pi + i \sin 4\pi)$... ??? ... yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $6e^{i\frac{11\pi}{6}}$??? $3\sqrt{3} - 3i$
 (f) $x^4 = 5$??? -5

1.



Braille letter

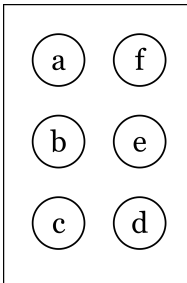
Complex Numbers 2, group *Iota* ι -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 4i|$??? $\sqrt{18}$
 (b) $2 + 2i$??? $\frac{\pi}{4}$
 (c) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$. ??? . no
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $7e^{i\frac{7\pi}{6}}$??? $-\frac{7\sqrt{3}}{2} + \frac{7}{2}i$
 (f) $x^3 = 4i$??? $4i$

2.



Braille letter

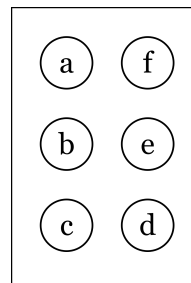
Complex Numbers 2, group *Iota* ι -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 5i|$??? $\sqrt{26}$
 (b) $5 - 5i$??? $\frac{7\pi}{4}$
 (c) $9(\cos 2\pi + i \sin 2\pi)$... ??? ... no
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $3e^{i\frac{3\pi}{4}}$??? $-\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$
 (f) $x^6 = 9$??? 9

3.



Braille letter

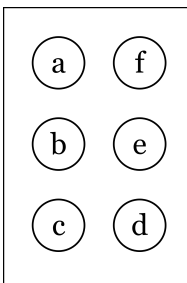
Complex Numbers 2, group *Iota* ι -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 3i|$??? $\sqrt{8}$
 (b) $8 - 8i$??? $\frac{7\pi}{4}$
 (c) $6(\cos \pi + i \sin \pi)$??? no
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $7e^{i\frac{3\pi}{4}}$??? $-\frac{7\sqrt{2}}{2} + \frac{7\sqrt{2}}{2}i$
 (f) $x^4 = 2i$??? $-2i$

4.



Braille letter

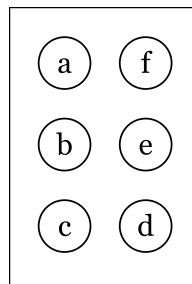
Complex Numbers 2, group *Kappa* κ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 2i|$??? 2
 (b) $2 - 2i$??? $\frac{7\pi}{4}$
 (c) $5(\cos 2\pi + i \sin 2\pi)$... ??? ... no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $9e^{i\frac{5\pi}{4}}$??? $-\frac{9\sqrt{2}}{2} + \frac{9\sqrt{2}}{2}i$
 (f) $x^5 = 4$??? 4

1.



Braille letter

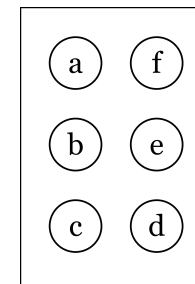
Complex Numbers 2, group *Kappa* κ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + i|$??? $\sqrt{5}$
 (b) $1 + i$??? $\frac{5\pi}{4}$
 (c) $5(\cos 2\pi + i \sin 2\pi)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $4e^{i\frac{4\pi}{3}}$??? $-4 - 4\sqrt{3}i$
 (f) $x^4 = 4$??? 4

2.



Braille letter

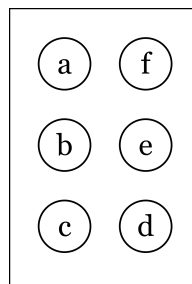
Complex Numbers 2, group *Kappa* κ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + 4i|$??? $\sqrt{52}$
 (b) $6 + 6i$??? $\frac{\pi}{4}$
 (c) $6(\cos 3\pi + i \sin 3\pi)$. ??? . no
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $3e^{i\frac{2\pi}{3}}$??? $-\frac{3}{2} - \frac{3\sqrt{3}}{2}i$
 (f) $x^4 = 5$??? $-5i$

3.



Braille letter

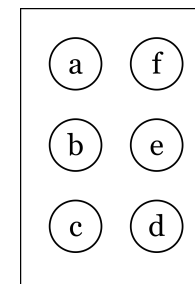
Complex Numbers 2, group *Kappa* κ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-7 + i|$??? $\sqrt{50}$
 (b) $-3 - 3i$??? $\frac{7\pi}{4}$
 (c) $6(\cos \pi + i \sin \pi)$??? yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $5e^{i\frac{\pi}{6}}$??? $\frac{5\sqrt{3}}{2} + \frac{5}{2}i$
 (f) $x^6 = -4$??? $4i$

4.



Braille letter

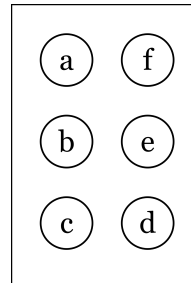
Complex Numbers 2, group *Lambda* λ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 4i|$??? $\sqrt{17}$
 (b) $-9 - 9i$??? $\frac{5\pi}{4}$
 (c) $6(\cos 2\pi + i \sin 2\pi)$??? yes
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3})$
 (e) $8e^{i\frac{3\pi}{4}}$??? $4\sqrt{2} + 4\sqrt{2}i$
 (f) $x^5 = -6i$??? -6

1.



Braille letter

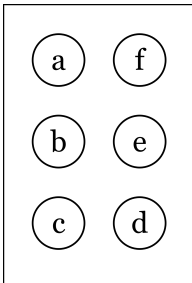
Complex Numbers 2, group *Lambda* λ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 3i|$??? $\sqrt{13}$
 (b) $9 - 9i$??? $\frac{\pi}{4}$
 (c) $2(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2})$... ??? ... no
 (d) $1 - \sqrt{3}i$??? $2(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6})$
 (e) $7e^{i\frac{4\pi}{3}}$??? $\frac{7}{2} - \frac{7\sqrt{3}}{2}i$
 (f) $x^3 = -8i$??? $8i$

2.



Braille letter

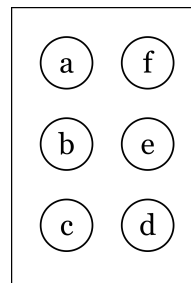
Complex Numbers 2, group *Lambda* λ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 4i|$??? $\sqrt{32}$
 (b) $-6 + 6i$??? $\frac{3\pi}{4}$
 (c) $6(\cos \pi + i \sin \pi)$??? no
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
 (e) $e^{i\frac{5\pi}{6}}$??? $-\frac{\sqrt{3}}{2} - \frac{1}{2}i$
 (f) $x^6 = 2$??? $-2i$

3.



Braille letter

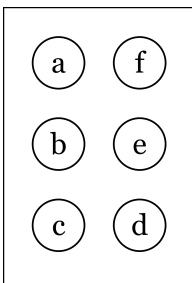
Complex Numbers 2, group *Lambda* λ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-8 + 5i|$??? $\sqrt{89}$
 (b) $1 + i$??? $\frac{\pi}{4}$
 (c) $2(\cos 2\pi + i \sin 2\pi)$. ??? . no
 (d) $1 + \sqrt{3}i$??? $2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
 (e) $5e^{i\frac{5\pi}{6}}$??? $\frac{5\sqrt{3}}{2} + \frac{5}{2}i$
 (f) $x^4 = -3$??? -3

4.



Braille letter

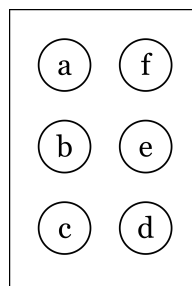
Complex Numbers 2, group $Mu \mu$ -i

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 6i|$??? $\sqrt{61}$
 (b) $5 + 5i$??? $\frac{\pi}{4}$
 (c) $2(\cos \pi + i \sin \pi)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $e^{i\frac{3\pi}{4}}$??? $-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
 (f) $x^6 = 4$??? -4

1.



Braille letter

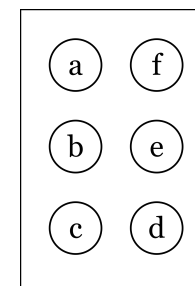
Complex Numbers 2, group $Mu \mu$ -ii

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 4i|$??? $\sqrt{33}$
 (b) $2 + 2i$??? $\frac{\pi}{4}$
 (c) $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $6e^{i\frac{2\pi}{3}}$??? $3 + 3\sqrt{3}i$
 (f) $x^3 = -4i$??? $-4i$

2.



Braille letter

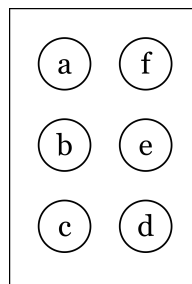
Complex Numbers 2, group $Mu \mu$ -iii

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|-8 + 2i|$??? $\sqrt{68}$
 (b) $-3 + 3i$??? $\frac{3\pi}{4}$
 (c) $2\left(\cos \frac{9\pi}{2} + i \sin \frac{9\pi}{2}\right)$... ??? ... yes
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $3e^{i\frac{3\pi}{4}}$??? $-\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$
 (f) $x^6 = 4$??? $-4i$

3.



Braille letter

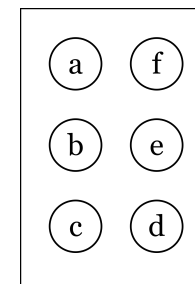
Complex Numbers 2, group $Mu \mu$ -iv

Name:

- (a) Calculate the absolute value of the complex number. (d) Write in the trigonometric form.
 (b) Determine the argument φ . (e) Write in the algebraic form.
 (c) Decide whether the complex number lies on the imaginary axis. (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + i|$??? $\sqrt{5}$
 (b) $7 + 7i$??? $\frac{\pi}{4}$
 (c) $6\left(\cos \frac{9\pi}{2} + i \sin \frac{9\pi}{2}\right)$... ??? ... yes
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $7e^{i\frac{5\pi}{6}}$??? $-\frac{7\sqrt{3}}{2} - \frac{7}{2}i$
 (f) $x^6 = -2i$??? $-2i$

4.



Braille letter

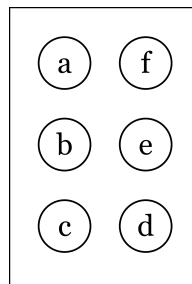
Complex Numbers 2, group $Nu \nu$ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + 7i|$??? $\sqrt{85}$
 (b) $4 - 4i$??? $\frac{\pi}{4}$
 (c) $4\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $e^{i\frac{\pi}{4}}$??? $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
 (f) $x^4 = 4$??? -4

1.



Braille letter

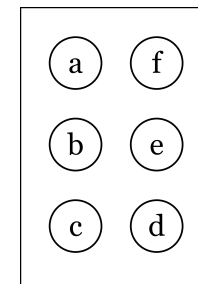
Complex Numbers 2, group $Nu \nu$ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 2i|$??? $\sqrt{29}$
 (b) $-6 - 6i$??? $\frac{7\pi}{4}$
 (c) $4\left(\cos \pi + i \sin \pi\right)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $7e^{i\frac{3\pi}{4}}$??? $-\frac{7\sqrt{2}}{2} - \frac{7\sqrt{2}}{2}i$
 (f) $x^4 = 3$??? 3

2.



Braille letter

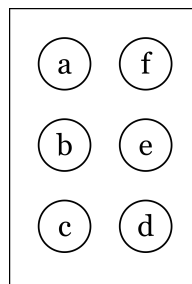
Complex Numbers 2, group $Nu \nu$ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 2i|$??? $\sqrt{8}$
 (b) $2 - 2i$??? $\frac{\pi}{4}$
 (c) $5\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $3e^{i\frac{7\pi}{6}}$??? $-\frac{3\sqrt{3}}{2} + \frac{3}{2}i$
 (f) $x^3 = 2$??? 2

3.



Braille letter

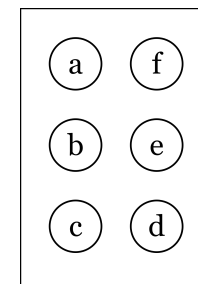
Complex Numbers 2, group $Nu \nu$ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - i|$??? $\sqrt{26}$
 (b) $8 + 8i$??? $\frac{5\pi}{4}$
 (c) $3\left(\cos 2\pi + i \sin 2\pi\right)$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $4e^{i\frac{4\pi}{3}}$??? $-2 - 2\sqrt{3}i$
 (f) $x^5 = 8$??? -8

4.



Braille letter

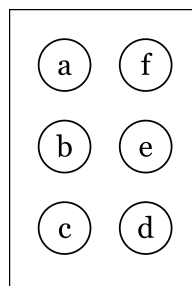
Complex Numbers 2, group $Xi \xi$ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 2i|$??? $\sqrt{29}$
 (b) $-5 - 5i$??? $\frac{5\pi}{4}$
 (c) $2(\cos 3\pi + i \sin 3\pi)$. ??? . yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $9e^{i\frac{11\pi}{6}}$??? $\frac{9\sqrt{3}}{2} - \frac{9}{2}i$
 (f) $x^4 = 2$??? $-2i$

1.



Braille letter

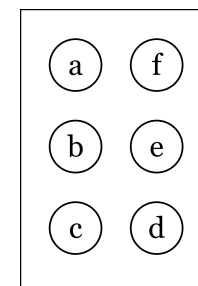
Complex Numbers 2, group $Xi \xi$ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 3i|$??? $\sqrt{10}$
 (b) $1 + i$??? $\frac{5\pi}{4}$
 (c) $4(\cos 2\pi + i \sin 2\pi)$... ??? ... no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $5e^{i\frac{3\pi}{4}}$??? $-\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$
 (f) $x^5 = -6i$??? -6

2.



Braille letter

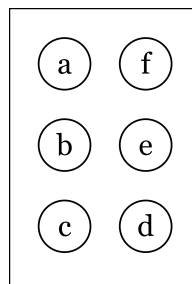
Complex Numbers 2, group $Xi \xi$ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|2 - 2i|$??? $\sqrt{8}$
 (b) $-4 + 4i$??? $\frac{3\pi}{4}$
 (c) $2(\cos \pi + i \sin \pi)$??? no
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $e^{i\frac{\pi}{4}}$??? $-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
 (f) $x^4 = -7i$??? $7i$

3.



Braille letter

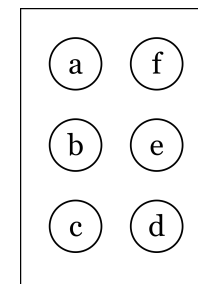
Complex Numbers 2, group $Xi \xi$ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 3i|$??? $\sqrt{13}$
 (b) $-7 - 7i$??? $\frac{7\pi}{4}$
 (c) $2\left(\cos \frac{9\pi}{2} + i \sin \frac{9\pi}{2}\right)$... ??? ... no
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $3e^{i\frac{\pi}{3}}$??? $\frac{3}{2} + \frac{3\sqrt{3}}{2}i$
 (f) $x^4 = 3$??? $-3i$

4.



Braille letter

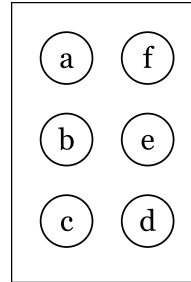
Complex Numbers 2, group *Omicron o* -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 6i|$??? $\sqrt{37}$
 (b) $-6 + 6i$??? $\frac{3\pi}{4}$
 (c) $9\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? no
 (d) $-3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
 (e) $3e^{i\frac{3\pi}{4}}$??? $-\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$
 (f) $x^4 = -5i$??? 5

1.



Braille letter

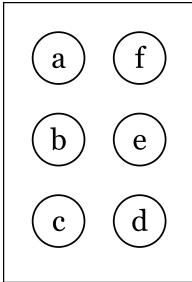
Complex Numbers 2, group *Omicron o* -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|8 - 5i|$??? $\sqrt{89}$
 (b) $-5 - 5i$??? $\frac{7\pi}{4}$
 (c) $2\left(\cos \frac{7\pi}{2} + i \sin \frac{7\pi}{2}\right)$... ??? ... no
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $7e^{i\frac{5\pi}{6}}$??? $\frac{7\sqrt{3}}{2} + \frac{7}{2}i$
 (f) $x^6 = 6i$??? 6i

2.



Braille letter

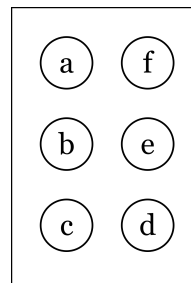
Complex Numbers 2, group *Omicron o* -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|7 - i|$??? $\sqrt{50}$
 (b) $-7 - 7i$??? $\frac{5\pi}{4}$
 (c) $4(\cos \pi + i \sin \pi)$??? no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $3e^{i\frac{\pi}{6}}$??? $-\frac{3\sqrt{3}}{2} + \frac{3}{2}i$
 (f) $x^3 = -2i$??? 2i

3.



Braille letter

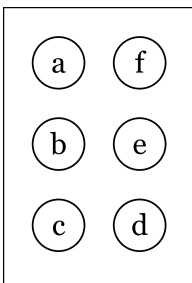
Complex Numbers 2, group *Omicron o* -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 7i|$??? $\sqrt{50}$
 (b) $1 + i$??? $\frac{\pi}{4}$
 (c) $3\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$??? no
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $7e^{i\frac{2\pi}{3}}$??? $\frac{7}{2} + \frac{7\sqrt{3}}{2}i$
 (f) $x^6 = 5$??? -5

4.



Braille letter

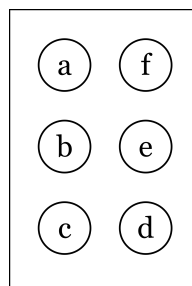
Complex Numbers 2, group $Pi \pi$ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|8 - 8i|$??? $\sqrt{128}$
 (b) $-1 + i$??? $\frac{5\pi}{4}$
 (c) $5(\cos 4\pi + i \sin 4\pi)$??? yes
 (d) $3 - \sqrt{3}i$??? $\sqrt{12} \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right)$
 (e) $2e^{i\frac{7\pi}{6}}$??? $-2\sqrt{3} - 2i$
 (f) $x^5 = 2i$??? $2i$

1.



Braille letter

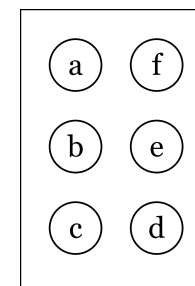
Complex Numbers 2, group $Pi \pi$ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 8i|$??? $\sqrt{80}$
 (b) $-3 - 3i$??? $\frac{5\pi}{4}$
 (c) $6(\cos \pi + i \sin \pi)$??? no
 (d) $-1 + \sqrt{3}i$??? $2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$
 (e) $6e^{i\frac{\pi}{4}}$??? $-3\sqrt{2} + 3\sqrt{2}i$
 (f) $x^5 = 3$??? $3i$

2.



Braille letter

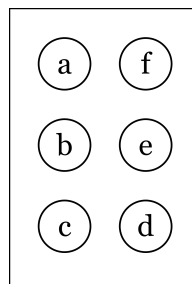
Complex Numbers 2, group $Pi \pi$ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 6i|$??? 4
 (b) $-4 - 4i$??? $\frac{5\pi}{4}$
 (c) $3(\cos \pi + i \sin \pi)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$
 (e) $8e^{i\frac{\pi}{6}}$??? $-4\sqrt{3} + 4i$
 (f) $x^3 = -5$??? -5

3.



Braille letter

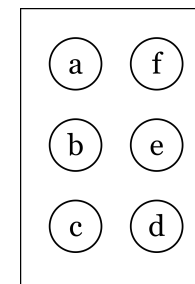
Complex Numbers 2, group $Pi \pi$ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 5i|$??? $\sqrt{29}$
 (b) $3 + 3i$??? $\frac{\pi}{4}$
 (c) $5 \left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2} \right)$... ??? ... yes
 (d) $1 - \sqrt{3}i$??? $2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right)$
 (e) $e^{i\frac{\pi}{6}}$??? $\sqrt{3} + i$
 (f) $x^5 = 6$??? 6

4.



Braille letter

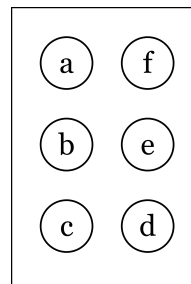
Complex Numbers 2, group *Rho* ρ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-3 + 5i|$??? $\sqrt{34}$
 (b) $-1 + i$??? $\frac{3\pi}{4}$
 (c) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$. ??? . yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $e^{i\frac{\pi}{4}}$??? $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
 (f) $x^3 = 4i$??? 4

1.



Braille letter

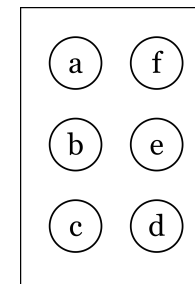
Complex Numbers 2, group *Rho* ρ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 4i|$??? $\sqrt{32}$
 (b) $-3 + 3i$??? $\frac{5\pi}{4}$
 (c) $3(\cos 2\pi + i \sin 2\pi)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $8e^{i\frac{2\pi}{3}}$??? $-8 + 8\sqrt{3}i$
 (f) $x^3 = 2$??? -2

2.



Braille letter

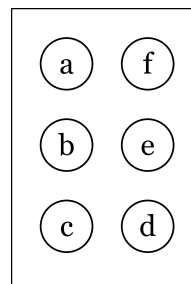
Complex Numbers 2, group *Rho* ρ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|5 - 7i|$??? $\sqrt{74}$
 (b) $4 + 4i$??? $\frac{\pi}{4}$
 (c) $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$??? yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $2e^{i\frac{\pi}{6}}$??? $\sqrt{3} - i$
 (f) $x^3 = -3$??? 3

3.



Braille letter

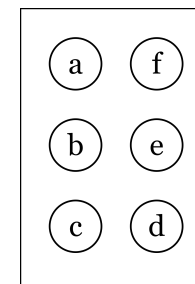
Complex Numbers 2, group *Rho* ρ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|2 - 7i|$??? $\sqrt{53}$
 (b) $-8 - 8i$??? $\frac{7\pi}{4}$
 (c) $7(\cos 2\pi + i \sin 2\pi)$... ??? ... no
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $9e^{i\frac{5\pi}{4}}$??? $-9\sqrt{2} - 9\sqrt{2}i$
 (f) $x^3 = 6$??? $6i$

4.



Braille letter

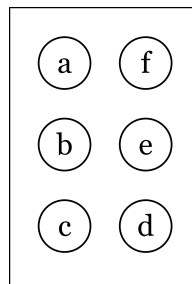
Complex Numbers 2, group *Sigma* σ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 8i|$??? $\sqrt{69}$
 (b) $-1 - i$??? $\frac{5\pi}{4}$
 (c) $6(\cos 2\pi + i \sin 2\pi)$... ??? ... no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $3e^{i\frac{3\pi}{4}}$??? $-\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$
 (f) $x^4 = -2i$??? $2i$

1.



Braille letter

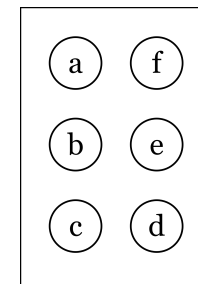
Complex Numbers 2, group *Sigma* σ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-3 + i|$??? $\sqrt{11}$
 (b) $2 + 2i$??? $\frac{\pi}{4}$
 (c) $3(\cos \pi + i \sin \pi)$??? yes
 (d) $-3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
 (e) $8e^{i\frac{5\pi}{3}}$??? $-4 - 4\sqrt{3}i$
 (f) $x^5 = -7$??? -7

2.



Braille letter

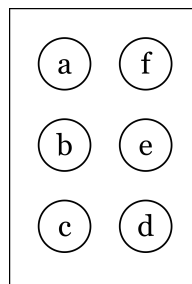
Complex Numbers 2, group *Sigma* σ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|3 - 6i|$??? $\sqrt{45}$
 (b) $-1 + i$??? $\frac{5\pi}{4}$
 (c) $2(\cos 3\pi + i \sin 3\pi)$??? no
 (d) $-3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
 (e) $e^{i\frac{3\pi}{4}}$??? $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
 (f) $x^3 = -7i$??? $-7i$

3.



Braille letter

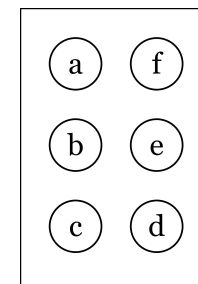
Complex Numbers 2, group *Sigma* σ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + 2i|$??? $\sqrt{40}$
 (b) $-1 - i$??? $\frac{7\pi}{4}$
 (c) $4\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$. ??? . no
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $2e^{i\frac{\pi}{6}}$??? $\sqrt{3} + i$
 (f) $x^5 = 6$??? -6

4.



Braille letter

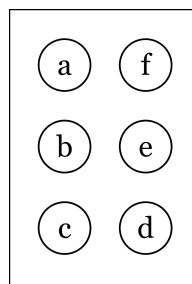
Complex Numbers 2, group *Tau* τ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 6i|$??? $\sqrt{37}$
 (b) $-3 - 3i$??? $\frac{7\pi}{4}$
 (c) $3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$.. ??? .. yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $e^{i\frac{5\pi}{6}}$??? $-\sqrt{3} + i$
 (f) $x^3 = -2i$??? $-2i$

1.



Braille letter

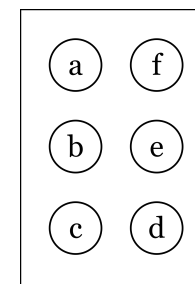
Complex Numbers 2, group *Tau* τ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 7i|$??? $\sqrt{53}$
 (b) $-5 + 5i$??? $\frac{5\pi}{4}$
 (c) $2\left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right)$... ??? ... no
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $4e^{i\frac{3\pi}{4}}$??? $-2\sqrt{2} + 2\sqrt{2}i$
 (f) $x^6 = -4i$??? 4

2.



Braille letter

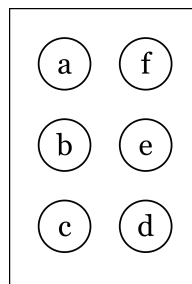
Complex Numbers 2, group *Tau* τ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 7i|$??? $\sqrt{53}$
 (b) $-5 + 5i$??? $\frac{5\pi}{4}$
 (c) $8(\cos \pi + i \sin \pi)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $4e^{i\frac{2\pi}{3}}$??? $-2 + 2\sqrt{3}i$
 (f) $x^3 = 4i$??? 4

3.



Braille letter

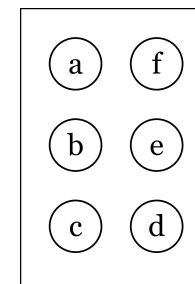
Complex Numbers 2, group *Tau* τ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 2i|$??? 1
 (b) $1 - i$??? $\frac{7\pi}{4}$
 (c) $2\left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right)$... ??? ... yes
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $e^{i\frac{3\pi}{4}}$??? $-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
 (f) $x^3 = -7i$??? $-7i$

4.



Braille letter

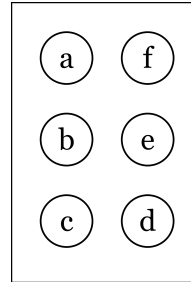
Complex Numbers 2, group *Upsilon v* -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|2 - 9i|$??? $\sqrt{85}$
 (b) $1 + i$??? $\frac{5\pi}{4}$
 (c) $6(\cos \pi + i \sin \pi)$??? yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $6e^{i\frac{11\pi}{6}}$??? $6\sqrt{3} - 6i$
 (f) $x^4 = 2$??? -2

1.



Braille letter

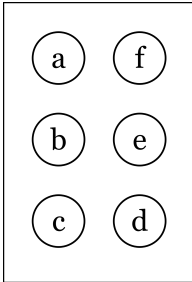
Complex Numbers 2, group *Upsilon v* -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-3 + i|$??? $\sqrt{8}$
 (b) $8 + 8i$??? $\frac{\pi}{4}$
 (c) $5\left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right)$??? no
 (d) $-3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
 (e) $e^{i\frac{7\pi}{6}}$??? $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
 (f) $x^3 = 7i$??? $7i$

2.



Braille letter

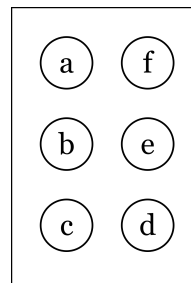
Complex Numbers 2, group *Upsilon v* -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|4 - 5i|$??? 1
 (b) $1 - i$??? $\frac{7\pi}{4}$
 (c) $2\left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right)$... ??? ... yes
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $6e^{i\frac{11\pi}{6}}$??? $3\sqrt{3} - 3i$
 (f) $x^3 = -3i$??? $-3i$

3.



Braille letter

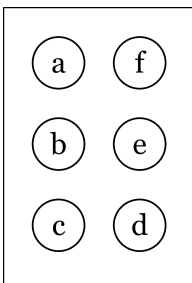
Complex Numbers 2, group *Upsilon v* -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|7 - 5i|$??? $\sqrt{74}$
 (b) $-5 - 5i$??? $\frac{7\pi}{4}$
 (c) $8\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$. ??? . yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $6e^{i\frac{4\pi}{3}}$??? $-3 - 3\sqrt{3}i$
 (f) $x^6 = -2i$??? $2i$

4.



Braille letter

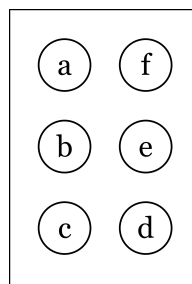
Complex Numbers 2, group *Phi* ϕ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|8 - 9i|$??? 1
 (b) $5 + 5i$??? $\frac{\pi}{4}$
 (c) $3\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... yes
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $4e^{i\frac{5\pi}{6}}$??? $-4\sqrt{3} + 4i$
 (f) $x^3 = 2$??? 2

1.



Braille letter

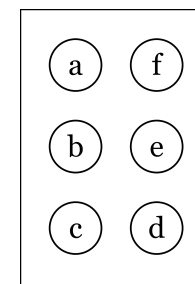
Complex Numbers 2, group *Phi* ϕ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|6 - i|$??? $\sqrt{37}$
 (b) $-1 + i$??? $\frac{3\pi}{4}$
 (c) $9(\cos 3\pi + i \sin 3\pi)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $3e^{i\frac{\pi}{3}}$??? $\frac{3}{2} + \frac{3\sqrt{3}}{2}i$
 (f) $x^4 = -6i$??? 6

2.



Braille letter

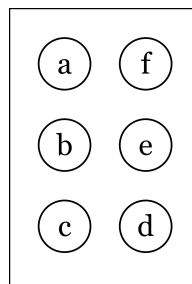
Complex Numbers 2, group *Phi* ϕ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|9 - i|$??? $\sqrt{82}$
 (b) $1 - i$??? $\frac{\pi}{4}$
 (c) $9(\cos 4\pi + i \sin 4\pi)$??? no
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $5e^{i\frac{\pi}{6}}$??? $\frac{5\sqrt{3}}{2} + \frac{5}{2}i$
 (f) $x^5 = -2i$??? $2i$

3.



Braille letter

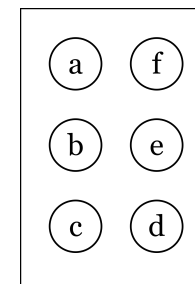
Complex Numbers 2, group *Phi* ϕ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 3i|$??? 5
 (b) $-6 + 6i$??? $\frac{3\pi}{4}$
 (c) $2(\cos 3\pi + i \sin 3\pi)$??? no
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $e^{i\frac{\pi}{6}}$??? $\frac{\sqrt{3}}{2} + \frac{1}{2}i$
 (f) $x^3 = -6i$??? -6

4.



Braille letter

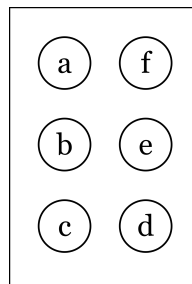
Complex Numbers 2, group *Chi* χ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 6i|$??? $\sqrt{37}$
 (b) $-3 - 3i$??? $\frac{7\pi}{4}$
 (c) $3(\cos 2\pi + i \sin 2\pi)$... ??? ... yes
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $4e^{i\frac{7\pi}{4}}$??? $4\sqrt{2} - 4\sqrt{2}i$
 (f) $x^4 = 4$??? -4

1.



Braille letter

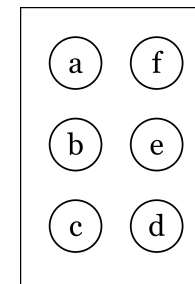
Complex Numbers 2, group *Chi* χ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|2 - 4i|$??? $\sqrt{20}$
 (b) $-3 - 3i$??? $\frac{7\pi}{4}$
 (c) $5(\cos 3\pi + i \sin 3\pi)$. ??? . yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $5e^{i\frac{11\pi}{6}}$??? $\frac{5\sqrt{3}}{2} + \frac{5}{2}i$
 (f) $x^3 = 2i$??? 2

2.



Braille letter

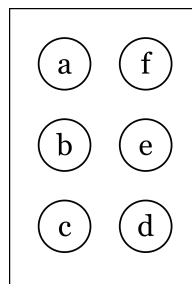
Complex Numbers 2, group *Chi* χ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|1 - 6i|$??? $\sqrt{35}$
 (b) $-8 + 8i$??? $\frac{3\pi}{4}$
 (c) $3(\cos 2\pi + i \sin 2\pi)$??? no
 (d) $-3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 (e) $2e^{i\frac{5\pi}{6}}$??? $\sqrt{3} + i$
 (f) $x^6 = -8i$??? $8i$

3.



Braille letter

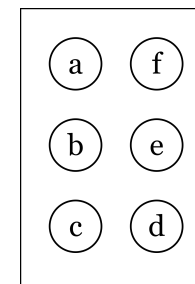
Complex Numbers 2, group *Chi* χ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-1 + 6i|$??? $\sqrt{37}$
 (b) $-2 - 2i$??? $\frac{5\pi}{4}$
 (c) $3\left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right)$??? no
 (d) $3 - \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$
 (e) $3e^{i\frac{7\pi}{6}}$??? $-\frac{3\sqrt{3}}{2} - \frac{3}{2}i$
 (f) $x^4 = -5$??? -5

4.



Braille letter

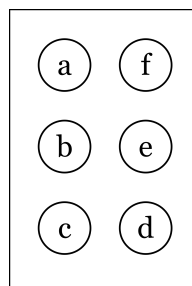
Complex Numbers 2, group *Psi* ψ -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + 3i|$??? $\sqrt{45}$
 (b) $1 + i$??? $\frac{5\pi}{4}$
 (c) $2(\cos \pi + i \sin \pi)$.. ??? .. yes
 (d) $1 + \sqrt{3}i$??? $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (e) $e^{i\frac{2\pi}{3}}$??? $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$
 (f) $x^3 = 2$??? 2

1.



Braille letter

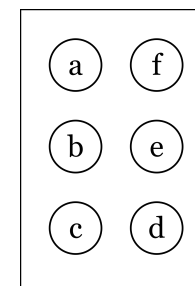
Complex Numbers 2, group *Psi* ψ -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + 6i|$??? $\sqrt{72}$
 (b) $4 - 4i$??? $\frac{\pi}{4}$
 (c) $8(\cos 3\pi + i \sin 3\pi)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $3e^{i\frac{7\pi}{4}}$??? $3\sqrt{2} - 3\sqrt{2}i$
 (f) $x^3 = -7i$??? -7

2.



Braille letter

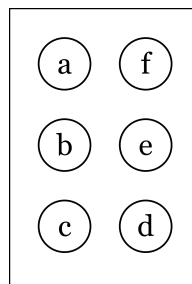
Complex Numbers 2, group *Psi* ψ -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-2 + 6i|$??? $\sqrt{32}$
 (b) $1 + i$??? $\frac{\pi}{4}$
 (c) $7\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... yes
 (d) $-1 - \sqrt{3}i$??? $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$
 (e) $6e^{i\frac{2\pi}{3}}$??? $-3 + 3\sqrt{3}i$
 (f) $x^6 = -4i$??? $4i$

3.



Braille letter

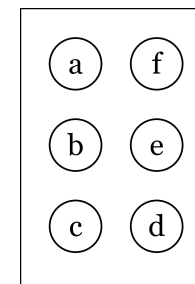
Complex Numbers 2, group *Psi* ψ -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + i|$??? $\sqrt{37}$
 (b) $1 + i$??? $\frac{5\pi}{4}$
 (c) $3\left(\cos \frac{7\pi}{2} + i \sin \frac{7\pi}{2}\right)$... ??? ... no
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $4e^{i\frac{5\pi}{4}}$??? $-2\sqrt{2} - 2\sqrt{2}i$
 (f) $x^5 = 5i$??? 5

4.



Braille letter

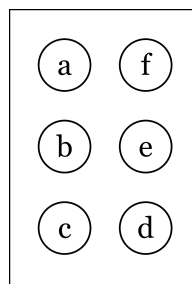
Complex Numbers 2, group *Omega* ω -i

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-7 + 3i|$??? $\sqrt{58}$
 (b) $-5 - 5i$??? $\frac{5\pi}{4}$
 (c) $5(\cos \pi + i \sin \pi)$??? yes
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $e^{i\frac{2\pi}{3}}$??? $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$
 (f) $x^4 = 3i$??? $-3i$

1.



Braille letter

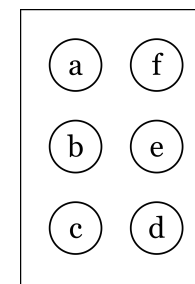
Complex Numbers 2, group *Omega* ω -ii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-6 + 4i|$??? $\sqrt{52}$
 (b) $2 - 2i$??? $\frac{\pi}{4}$
 (c) $3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$... ??? ... yes
 (d) $3 + \sqrt{3}i$??? $\sqrt{12}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
 (e) $e^{i\frac{5\pi}{3}}$??? $\frac{1}{2} - \frac{\sqrt{3}}{2}i$
 (f) $x^3 = -5$??? $-5i$

2.



Braille letter

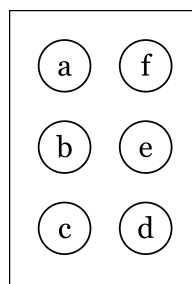
Complex Numbers 2, group *Omega* ω -iii

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|9 - 3i|$??? $\sqrt{90}$
 (b) $-1 - i$??? $\frac{7\pi}{4}$
 (c) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$... ??? ... no
 (d) $1 - \sqrt{3}i$??? $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$
 (e) $5e^{i\frac{2\pi}{3}}$??? $\frac{5}{2} + \frac{5\sqrt{3}}{2}i$
 (f) $x^6 = 5i$??? -5

3.



Braille letter

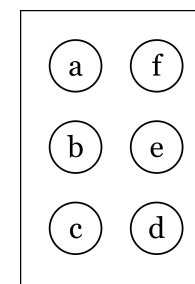
Complex Numbers 2, group *Omega* ω -iv

Name:

- (a) Calculate the absolute value of the complex number.
 (b) Determine the argument φ .
 (c) Decide whether the complex number lies on the imaginary axis.
- (d) Write in the trigonometric form.
 (e) Write in the algebraic form.
 (f) Solve the equation and calculate the product of its solutions.

- (a) $|-4 + 2i|$??? $\sqrt{20}$
 (b) $3 - 3i$??? $\frac{7\pi}{4}$
 (c) $9(\cos 2\pi + i \sin 2\pi)$... ??? ... no
 (d) $-1 + \sqrt{3}i$??? $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 (e) $6e^{i\frac{5\pi}{3}}$??? $-3 - 3\sqrt{3}i$
 (f) $x^4 = -2$??? $2i$

4.



Braille letter

Complex Numbers 2 (solutions)

ν	$i : \mathbf{N}$	(a) $\sqrt{85}\checkmark$	(b) $\frac{7\pi}{4}\times$	(c) yes $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{11\pi}{6}+i\sin\frac{11\pi}{6}\right)\times$	(e) $\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\checkmark$	(f) $-4\checkmark$
	$ii : \mathbf{A}$	(a) $\sqrt{29}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-\frac{7\sqrt{2}}{2}+\frac{7\sqrt{2}}{2}i\times$	(f) $-3\times$
	$iii : \mathbf{M}$	(a) $\sqrt{8}\checkmark$	(b) $\frac{7\pi}{4}\times$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-\frac{3\sqrt{3}}{2}-\frac{3}{2}i\times$	(f) $2\checkmark$
	$iv : \mathbf{E}$	(a) $\sqrt{26}\checkmark$	(b) $\frac{\pi}{4}\times$	(c) no $\times$	(d) $\sqrt{12}\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)\times$	(e) $-2-2\sqrt{3}i\checkmark$	(f) $8\times$
	ξ	$i : \mathbf{H}$	(a) $\sqrt{29}\checkmark$	(b) $\frac{5\pi}{4}\checkmark$	(c) no $\times$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\times$	(e) $\frac{9\sqrt{3}}{2}-\frac{9}{2}i\checkmark$
ο	$ii : \mathbf{O}$	(a) $\sqrt{10}\checkmark$	(b) $\frac{\pi}{4}\times$	(c) no $\checkmark$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $-\frac{5\sqrt{2}}{2}+\frac{5\sqrt{2}}{2}i\checkmark$	(f) $-6i\times$
	$iii : \mathbf{P}$	(a) $\sqrt{8}\checkmark$	(b) $\frac{3\pi}{4}\checkmark$	(c) no $\checkmark$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\times$	(f) $7i\checkmark$
	$iv : \mathbf{E}$	(a) $\sqrt{13}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $\frac{3}{2}+\frac{3\sqrt{3}}{2}i\checkmark$	(f) $-3\times$
	$i : \mathbf{H}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{3\pi}{4}\checkmark$	(c) yes $\times$	(d) $\sqrt{12}\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)\times$	(e) $-\frac{3\sqrt{2}}{2}+\frac{3\sqrt{2}}{2}i\checkmark$	(f) $5i\times$
	$ii : \mathbf{A}$	(a) $\sqrt{89}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $-\frac{7\sqrt{3}}{2}+\frac{7}{2}i\times$	(f) $-6i\times$
π	$iii : \mathbf{L}$	(a) $\sqrt{50}\checkmark$	(b) $\frac{5\pi}{4}\checkmark$	(c) no $\checkmark$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $\frac{3\sqrt{3}}{2}+\frac{3}{2}i\times$	(f) $-2i\times$
	$iv : \mathbf{F}$	(a) $\sqrt{50}\checkmark$	(b) $\frac{\pi}{4}\checkmark$	(c) yes $\times$	(d) $\sqrt{12}\left(\cos\frac{7\pi}{6}+i\sin\frac{7\pi}{6}\right)\times$	(e) $-\frac{7}{2}+\frac{7\sqrt{3}}{2}i\times$	(f) $-5\checkmark$
	$i : \mathbf{C}$	(a) $\sqrt{128}\checkmark$	(b) $\frac{3\pi}{4}\times$	(c) no $\times$	(d) $\sqrt{12}\left(\cos\frac{11\pi}{6}+i\sin\frac{11\pi}{6}\right)\times$	(e) $-\sqrt{3}-i\times$	(f) $2i\checkmark$
	$ii : \mathbf{L}$	(a) $\sqrt{80}\checkmark$	(b) $\frac{4}{4}\checkmark$	(c) no $\checkmark$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $3\sqrt{2}+3\sqrt{2}i\times$	(f) $3\times$
	$iii : \mathbf{I}$	(a) $\sqrt{40}\times$	(b) $\frac{5\pi}{4}\checkmark$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $4\sqrt{3}+4i\times$	(f) $-5\checkmark$
ρ	$iv : \mathbf{P}$	(a) $\sqrt{29}\checkmark$	(b) $\frac{\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $\frac{\sqrt{3}}{2}+\frac{1}{2}i\times$	(f) $6\checkmark$
	$i : \mathbf{W}$	(a) $\sqrt{34}\checkmark$	(b) $\frac{3\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\checkmark$	(e) $\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\checkmark$	(f) $4i\times$
	$ii : \mathbf{A}$	(a) $\sqrt{32}\checkmark$	(b) $\frac{3\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-4+4\sqrt{3}i\times$	(f) $2\times$
	$iii : \mathbf{L}$	(a) $\sqrt{74}\checkmark$	(b) $\frac{\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $\sqrt{3}+i\times$	(f) $-3\times$
	$iv : \mathbf{K}$	(a) $\sqrt{53}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) no $\checkmark$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $-\frac{9\sqrt{2}}{2}-\frac{9\sqrt{2}}{2}i\times$	(f) $6\times$
σ	$i : \mathbf{T}$	(a) $\sqrt{68}\times$	(b) $\frac{5\pi}{4}\checkmark$	(c) no $\checkmark$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $-\frac{3\sqrt{2}}{2}+\frac{3\sqrt{2}}{2}i\checkmark$	(f) $2i\checkmark$
	$ii : \mathbf{I}$	(a) $\sqrt{10}\times$	(b) $\frac{\pi}{4}\checkmark$	(c) no $\times$	(d) $\sqrt{12}\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)\times$	(e) $4-4\sqrt{3}i\times$	(f) $-7\checkmark$
	$iii : \mathbf{M}$	(a) $\sqrt{45}\checkmark$	(b) $\frac{3\pi}{4}\times$	(c) no $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)\times$	(e) $-\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\times$	(f) $-7i\checkmark$
	$iv : \mathbf{E}$	(a) $\sqrt{40}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\times$	(e) $\sqrt{3}+i\checkmark$	(f) $6\times$
	$i : \mathbf{M}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\times$	(e) $-\frac{\sqrt{3}}{2}+\frac{1}{2}i\times$	(f) $-2i\checkmark$
τ	$ii : \mathbf{E}$	(a) $\sqrt{53}\checkmark$	(b) $\frac{3\pi}{4}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-2\sqrt{2}+2\sqrt{2}i\checkmark$	(f) $4i\times$
	$iii : \mathbf{E}$	(a) $\sqrt{53}\checkmark$	(b) $\frac{3\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-2+2\sqrt{3}i\checkmark$	(f) $4i\times$
	$iv : \mathbf{T}$	(a) $\sqrt{5}\times$	(b) $\frac{7\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{11\pi}{6}+i\sin\frac{11\pi}{6}\right)\times$	(e) $-\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\checkmark$	(f) $-7i\checkmark$
	$i : \mathbf{C}$	(a) $\sqrt{85}\checkmark$	(b) $\frac{\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $3\sqrt{3}-3i\times$	(f) $-2\checkmark$
	$ii : \mathbf{I}$	(a) $\sqrt{10}\times$	(b) $\frac{\pi}{4}\checkmark$	(c) yes $\times$	(d) $\sqrt{12}\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)\times$	(e) $-\frac{\sqrt{3}}{2}-\frac{1}{2}i\times$	(f) $7i\checkmark$
υ	$iii : \mathbf{T}$	(a) $\sqrt{41}\times$	(b) $\frac{7\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $3\sqrt{3}-3i\checkmark$	(f) $-3i\checkmark$
	$iv : \mathbf{Y}$	(a) $\sqrt{74}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\checkmark$	(e) $-3-3\sqrt{3}i\checkmark$	(f) $2i\checkmark$
	$i : \mathbf{S}$	(a) $\sqrt{145}\times$	(b) $\frac{\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $-2\sqrt{3}+2i\times$	(f) $2\checkmark$
	$ii : \mathbf{H}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{3\pi}{4}\checkmark$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $\frac{3}{2}+\frac{3\sqrt{3}}{2}i\checkmark$	(f) $6i\times$
	$iii : \mathbf{O}$	(a) $\sqrt{82}\checkmark$	(b) $\frac{7\pi}{4}\times$	(c) no $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{7\pi}{6}+i\sin\frac{7\pi}{6}\right)\times$	(e) $\frac{5\sqrt{3}}{2}+\frac{5}{2}i\checkmark$	(f) $-2i\times$
φ	$iv : \mathbf{W}$	(a) $5\checkmark$	(b) $\frac{3\pi}{4}\checkmark$	(c) no $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{7\pi}{6}+i\sin\frac{7\pi}{6}\right)\checkmark$	(e) $\frac{\sqrt{3}}{2}+\frac{1}{2}i\checkmark$	(f) $-6i\times$
	$i : \mathbf{C}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $2\sqrt{2}-2\sqrt{2}i\times$	(f) $-4\checkmark$
	$ii : \mathbf{A}$	(a) $\sqrt{20}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\times$	(e) $\frac{5\sqrt{3}}{2}-\frac{5}{2}i\times$	(f) $2i\times$
	$iii : \mathbf{S}$	(a) $\sqrt{37}\times$	(b) $\frac{3\pi}{4}\checkmark$	(c) no $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{7\pi}{6}+i\sin\frac{7\pi}{6}\right)\times$	(e) $-\sqrt{3}+i\times$	(f) $8i\checkmark$
	$iv : \mathbf{H}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{5\pi}{4}\checkmark$	(c) yes $\times$	(d) $\sqrt{12}\left(\cos\frac{11\pi}{6}+i\sin\frac{11\pi}{6}\right)\times$	(e) $-\frac{3\sqrt{3}}{2}-\frac{3}{2}i\checkmark$	(f) $5\times$
ψ	$i : \mathbf{D}$	(a) $\sqrt{45}\checkmark$	(b) $\frac{\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\times$	(e) $-\frac{1}{2}+\frac{\sqrt{3}}{2}i\checkmark$	(f) $2\checkmark$
	$ii : \mathbf{A}$	(a) $\sqrt{72}\checkmark$	(b) $\frac{7\pi}{4}\times$	(c) no $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $\frac{3\sqrt{2}}{2}-\frac{3\sqrt{2}}{2}i\times$	(f) $-7i\times$
	$iii : \mathbf{T}$	(a) $\sqrt{40}\times$	(b) $\frac{\pi}{4}\checkmark$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-3+3\sqrt{3}i\checkmark$	(f) $4i\checkmark$
	$iv : \mathbf{E}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{\pi}{4}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $-2\sqrt{2}-2\sqrt{2}i\checkmark$	(f) $5i\times$
	$i : \mathbf{G}$	(a) $\sqrt{58}\checkmark$	(b) $\frac{5\pi}{4}\checkmark$	(c) no $\times$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $-\frac{1}{2}+\frac{\sqrt{3}}{2}i\checkmark$	(f) $-3i\checkmark$
ω	$ii : \mathbf{O}$	(a) $\sqrt{52}\checkmark$	(b) $\frac{7\pi}{4}\times$	(c) yes $\checkmark$	(d) $\sqrt{12}\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)\times$	(e) $\frac{1}{2}-\frac{\sqrt{3}}{2}i\checkmark$	(f) $-5\times$
	$iii : \mathbf{A}$	(a) $\sqrt{90}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}\right)\times$	(e) $-\frac{5}{2}+\frac{5\sqrt{3}}{2}i\checkmark$	(f) $-5i\times$
	$iv : \mathbf{L}$	(a) $\sqrt{20}\checkmark$	(b) $\frac{7\pi}{4}\checkmark$	(c) no $\checkmark$	(d) $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)\times$	(e) $3-3\sqrt{3}i\times$	(f) $2\times$
	$i : \mathbf{M}$	(a) $\sqrt{37}\checkmark$	(b) $\frac{5\pi}{4}\times$	(c) yes $\checkmark$	(d) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\times$	(e) $-\frac{\sqrt{3}}{2}+\frac{1}{2}i\checkmark$	(f) $-2i\checkmark$
	$ii : \mathbf{E}$	(a) $\sqrt{53}\checkmark$	(b) $\frac{\pi}{3}\times$	(c) yes $\times$	(d) $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\times$	(e) $-2\sqrt{2}+2\sqrt{2}i\checkmark$	(f) $4i\times$